Harvard John A. Paulson School of Engineering and Applied Sciences

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Introduction & Related work

Challenges in LLMs

- Scaling up model size alone has not proved sufficient for achieving high performance on challenging tasks, such as arithmetic, commonsense, and symbolic reasoning.
- Large language models still have limitations in their ability to reason and understand the context of a situation.

Previous work

- Models have the ability to generate neutral language intermediate steps by training from scratch (Ling et al., 2017) or finetuning a pretrained model (Cobbe et al., 2021).
    - costly to create a large set of high quality rationales
- Large language models offer the exciting prospect of in-context few-shot learning via *prompting* (Brown et al., 2020)
    - works poorly on tasks that require reasoning abilities,
    - often does not improve substantially with increasing language model scale

Contribution

- This work explores the ability of language models to perform few-shot prompting for reasoning tasks, given a prompt that consists of triples: ⟨input, *chain of thought*, output⟩.
    - Chain-of-thought: a series of intermediate natural language reasoning steps that lead to the final output (*chain-of-thought prompting*).
- This work also presents empirical evaluations on arithmetic, commonsense, and symbolic reasoning benchmarks, showing that chain-of-thought prompting outperforms standard prompting.

Methodology

How do we solve complicated problems

Decompose the problem into intermediate steps and solve each before giving the final answer

"After Jane gives 2 flowers to her mom she has 10 . . . then after she gives 3 to her dad she will have 7 . . . so the answer is 7."

The paper shows sufficiently large LMs can generate chains of thought if demonstrations of chain-of-thought reasoning are provided as few-shot prompting.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✔

Benefits

1. Allows models to decompose multi-step problems into intermediate steps so additional computation can be allocated to problems that require more reasoning steps.
2. Interpretable method to understand model's response
3. Potentially applicable to any task that require multi-step approach.
4. Can be induced simply by including examples of chain of thought sequences while prompting.

Sensitivity to prompt engineering

Chain of thought is robust to:

1. Different annotators writing prompts
2. Annotators without machine learning background writing prompts
3. Different types, ordering and number of prompt examples
4. Different language models (LaMDA, GPT-3, and PaLM)

However Prompt Engineering improved performance significantly in many cases.

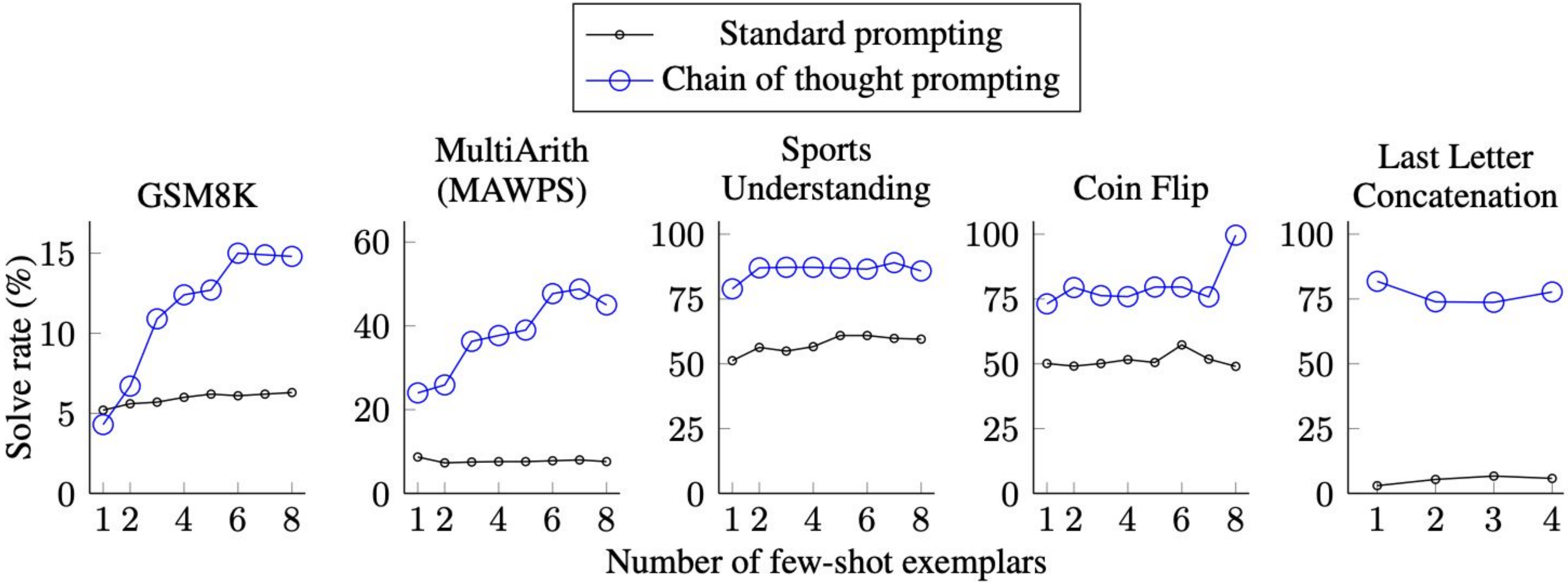

Figure 11: The improvement of chain of thought prompting over standard prompting appears robust to varying the number of few-shot exemplars in the prompt.

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Experiments & Results

Experimental Setup

Benchmarks: five math word problem benchmarks:

- GSM8K (Cobbe et al., 2021)
- SVAMP (Patel et al., 2021)
- ASDiv (Miao et al., 2020)
- AQuA
- MAWPS (Koncel-Kedziorski et al., 2016)

PROMPT FOR MATH WORD PROBLEMS

Q: There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done, there will be 21 trees. How many trees did the grove workers plant today?

A: There are 15 trees originally. Then there were 21 trees after some more were planted. So there must have been 21 - 15 = 6. The answer is 6.

Experimental Setup

Models:

- baseline: standard few-shot prompting (Brown et al., 2020)
- chain-of-thought prompting: augment each exemplar in few-shot prompting with a chain of thought for an associated answer, using five different large language models:
    - GPT-3 (Brown et al., 2020)
    - LaMDA (Thoppilan et al., 2022)
    - PaLM
    - UL2 20B (Tay et al., 2022)
    - Codex (Chen et al., 2021)

Results

Three key takeaways:

- chain-of-thought prompting is an emergent ability of **model scale**;
- chain-of-thought prompting has larger performance gains for **more-complicated problems**;
- chain-of-thought prompting via GPT-3 175B and PaLM 540B **compares favorably** to prior state of the art.

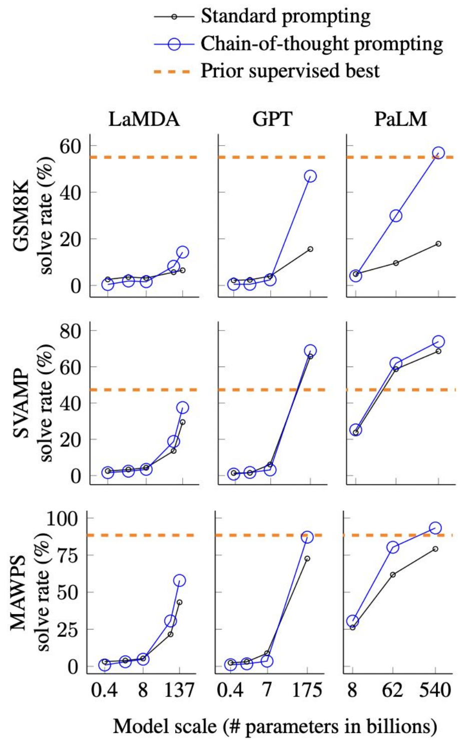


Figure 4: Chain-of-thought prompting enables large language models to solve challenging math problems. Notably, chain-of-thought reasoning is an emergent ability of increasing model scale.

Ablation Study

An ablation study with three variations of chain of thought:

- Equation only
- Variable compute only
- Chain of thought after answer

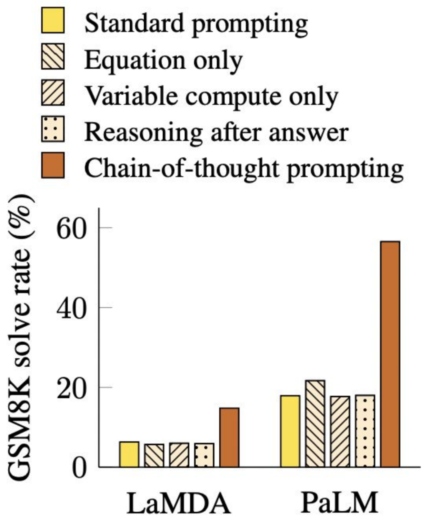


Figure 5: Ablation study for different variations of prompting using LaMDA 137B and PaLM 540B. Results for other datasets are given in Appendix Table 6 and Table 7.

Robustness of Chain of Thought

Findings: chain-of-thought prompting for arithmetic reasoning is **robust** to

- annotators
- independently-written chains of thought
- different exemplars
- various language models
- different exemplar orders
- varying numbers of exemplars

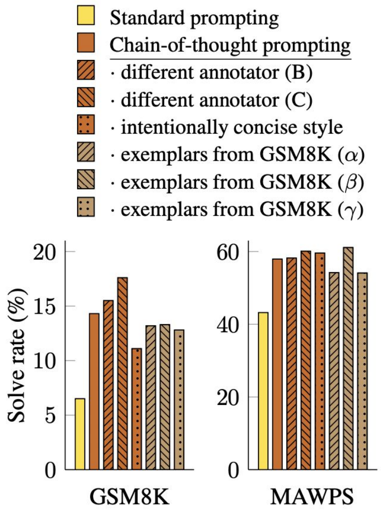


Figure 6: Chain-of-thought prompting has variance for different prompt examples (as expected) but outperforms standard prompting for various annotators as well as for different exemplars.

Commonsense Reasoning

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. 9 + 90(2) + 401(3) = 1392. The answer is (b).

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.
Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

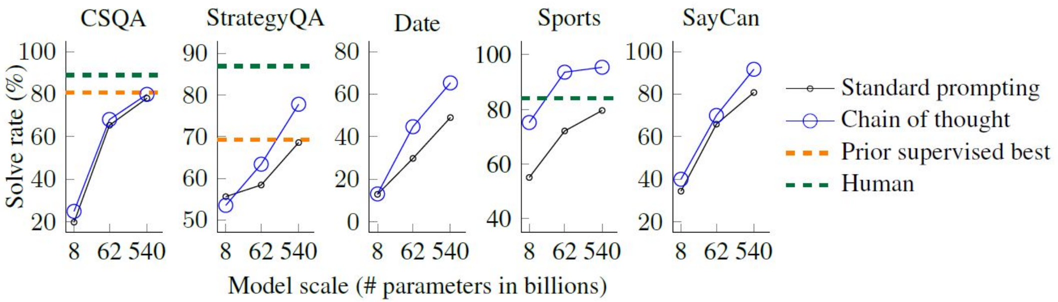
A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Results



- Chain of thought prompting works better when the model is sufficiently large.

Table 4: Standard prompting versus chain of thought prompting on five commonsense reasoning benchmarks. Chain of thought prompting is an emergent ability of model scale—it does not positively impact performance until used with a model of sufficient scale.

		CSQA		StrategyQA		Date		Sports		SayCan	
Model		standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT
UL2	20B	34.2	**51.4**	59.0	53.3	13.5	**14.0**	57.9	**65.3**	20.0	**41.7**
LaMDA	420M	20.1	19.2	46.4	24.9	1.9	1.6	50.0	49.7	7.5	7.5
	2B	20.2	19.6	52.6	45.2	8.0	6.8	49.3	57.5	8.3	8.3
	8B	19.0	20.3	54.1	46.8	9.5	5.4	50.0	52.1	28.3	33.3
	68B	37.0	**44.1**	59.6	**62.2**	15.5	**18.6**	55.2	**77.5**	35.0	**42.5**
	137B	53.6	**57.9**	62.4	**65.4**	21.5	**26.8**	59.5	**85.8**	43.3	**46.6**
GPT	350M	14.7	15.2	20.6	0.9	4.3	0.9	33.8	41.6	12.5	0.8
	1.3B	12.0	19.2	45.8	35.7	4.0	1.4	0.0	26.9	20.8	9.2
	6.7B	19.0	**24.0**	53.6	50.0	8.9	4.9	0.0	4.4	17.5	**35.0**
	175B	79.5	73.5	65.9	65.4	43.8	**52.1**	69.6	**82.4**	81.7	**87.5**
Codex	-	82.3	77.9	67.1	**73.2**	49.0	**64.8**	71.7	**98.5**	85.8	**88.3**
PaLM	8B	19.8	**24.9**	55.6	53.5	12.9	13.1	55.1	**75.2**	34.2	**40.0**
	62B	65.4	**68.1**	58.4	**63.4**	29.8	**44.7**	72.1	**93.6**	65.8	**70.0**
	540B	78.1	**79.9**	68.6	**77.8**	49.0	**65.3**	80.5	**95.4**	80.8	**91.7**

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Symbolic Reasoning

Math Word Problems (free response)

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Results

- In domain and out-of-domain evaluations

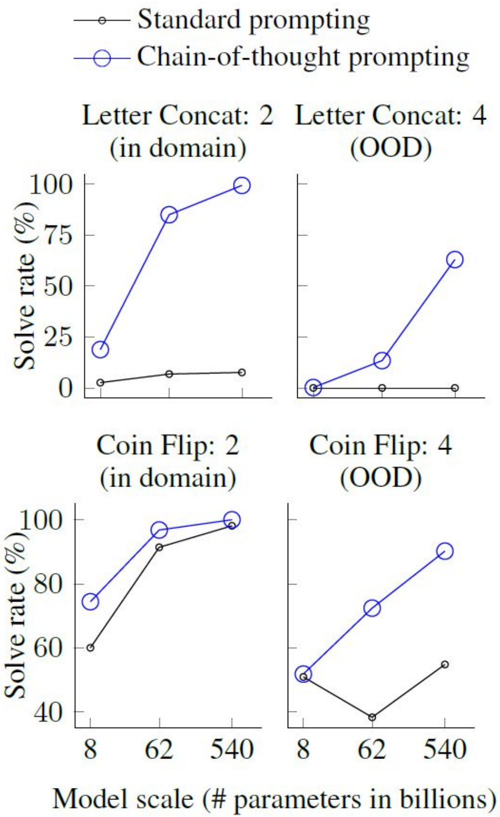

Table 5: Standard prompting versus chain of thought prompting enables length generalization to longer inference examples on two symbolic manipulation tasks.

		Last Letter Concatenation						Coin Flip (state tracking)					
		2		OOD: 3		OOD: 4		2		OOD: 3		OOD: 4	
Model		standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT
UL2	20B	0.6	**18.8**	0.0	0.2	0.0	0.0	70.4	67.1	51.6	52.2	48.7	50.4
LaMDA	420M	0.3	**1.6**	0.0	0.0	0.0	0.0	52.9	49.6	50.0	50.5	49.5	49.1
	2B	2.3	**6.0**	0.0	0.0	0.0	0.0	54.9	**55.3**	47.4	48.7	49.8	50.2
	8B	1.5	**11.5**	0.0	0.0	0.0	0.0	52.9	**55.5**	48.2	49.6	51.2	50.6
	68B	4.4	**52.0**	0.0	**0.8**	0.0	**2.5**	56.2	**83.2**	50.4	**69.1**	50.9	**59.6**
	137B	5.8	**77.5**	0.0	**34.4**	0.0	**13.5**	49.0	**99.6**	50.7	**91.0**	49.1	**74.5**
PaLM	8B	2.6	**18.8**	0.0	0.0	0.0	**0.2**	60.0	**74.4**	47.3	**57.1**	50.9	**51.8**
	62B	6.8	**85.0**	0.0	**59.6**	0.0	**13.4**	91.4	**96.8**	43.9	**91.0**	38.3	**72.4**
	540B	7.6	**99.4**	0.2	**94.8**	0.0	**63.0**	98.1	**100.0**	49.3	**98.6**	54.8	**90.2**

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Limitations & Discussion

Limitations:

- Although chain of thought emulates the thought processes of human reasoners, this does not answer whether the neural network is actually “reasoning.
- The emergence of chain-of-thought reasoning only at large model scales makes it costly to serve in real-world applications.

Discussion:

- What are some potential limitations of using chain-of-thought prompting for large language models? Are there certain types of reasoning tasks that may not be well-suited to this approach?
- What other prompting methods might expand the range of tasks that language models can solve?
- The authors argue that chain-of-thought prompting has the potential to improve the interpretability of language models. Do you agree with this claim and why?

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Thanks & Questions

Can language models learn from explanations in context?

Andrew Lampinen, Ishita Dasgupta, Stephanie Chan, Kory Matthewson, Michael Henry Tessler, Antonia Creswell, James McClelland, Jane Wang, Felix Hill

DeepMind, London, UK

Presented by Cynthia Chen, Jason Jabbour, Mark Mazumder

April 17, 2023

Presentation Roadmap

- Introduction
- Method
- Experiments
- Discussion

Introduction

- **Language models (LMs)** have been found to be able to perform new tasks by adapting to a few in-context examples
    - In-context example: an example (question + right answer) for a specific task prompt
    - “few-shot”: a few examples are given to guide the model

- Could including few-shot **explanations** of the answers in these examples improve LM performance?

Training: Task instruction, examples + explanations

Task instruction

Answer these questions by identifying whether the second sentence is an appropriate paraphrase of the first, metaphorical sentence.

Few-shot example #1

Q: David's eyes were like daggers at Paul when Paul invited his new girlfriend to dance. <- -> David had two daggers when Paul invited his new girlfriend to dance.
choice: True
choice: False
A: False

Answer explanation

Explanation: David's eyes were not literally daggers, it is a metaphor used to imply that David was glaring fiercely at Paul.

4 more examples
+ explanations

⋮

Evaluation: Target question

Target question

Q: Our whole life we swim against the waves towards the green light of happiness. <- -> Our whole life we try to reach happiness.
choice: True
choice: False
A:

Related Work:

- In-context and prompt-based learning:
    - **Min et al., 2022**: Find that ground truth labels don't have a large effect on model performance, and identify other aspects (label space / distribution) that drive performance
    - **Webson and Pavlick, 2021**: Find limitation in models' ability to truly understand the meaning of their prompts
- Prompting with explicit instructions
    - **Liu et al., 2021**: Prompting with explicit instructions or task descriptions helps LMs adapt to a task
    - **Wei et al, 2022**: Breaking down the steps of a reasoning process for LMs improve few-shot performance
- Exploring explanations of answers
    - Assessing effects of auxiliary in-context information on performance
    - Large body of work (beyond prompting) on training/tuning with explanations
    - This paper particularly focuses on the effect of **post-answer explanations**

Research Questions

- Can language models benefit from explanations when learning from **examples in-context**?

- Do **few-shot explanations** help the model to **“understand” the task better**? Do explanations of answers improve few-shot task performance?

- More generally, what kind of **in-context learning abilities** do LMs exhibit?

Contributions and Main Findings

- Annotate 40 challenging tasks with explanations of examples
- Evaluate the particular **effect of post-answer explanations** (contrasted to previous work such as Wei et al. who provide chains of reasoning *before* the answer)
- Findings:
    - **Explanations of examples improve the performance of LLMs** when compared to matched control conditions
    - Explanations tuned using a small validation set have even larger performance benefits

Presentation Roadmap

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Methods: Data and Model

- **Data**: Selected a set of **40 challenging tasks** from the BIG-Bench dataset
  - Task examples:
    - Inferring goals from actions
    - Reasoning about mathematical induction
    - Reasoning about causality
    - Inferring assumptions behind a statement
- **Model**: set of decoder-only Transformer language models
  - Evaluated a set of models with same context window and trained on the same dataset

Methods: Explanation Annotation

- An author **annotated 15 examples with human explanations**
  - Restricted to multiple choice tasks
- Conducted comparison to ***control explanations*** to account for confounding factors
  - Scrambled explanations
  - True non-explanations
  - Permuted explanations
- **Explanation fine-tuning**
  - Selected examples to build a 5-shot prompt greedily, to evaluate the benefit of selecting the best examples
  - Hand-tuned explanations to better understand the potential benefits of more optimal “expert” explanations
  - Tuning performed on small validation set, but tested on larger set of task examples

Methods: Evaluation

- **Modeled dependencies of results using hierarchical logistic regression**
  - Nested dependencies and heterogeneous structure
  - Factors to take into account:
    - Task difficulty
    - Shared content between prompts
- Model each unique effect at each level of the hierarchy
  - Estimates the unique added contribution of a prompt component (eg: few-shot example, explanation, etc) to the performance

Presentation Roadmap

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Walkthrough → **Potential Impacts** → **Nuances**

Benefit of Adding Explanatio

Look at **2 out of 15 prompts** drawn from the 40 tasks

analytic_entailment

odd_one_out

physical_intuition

unit_interpretation/lv2

riddle_sense

arithmetic/3_digit_division

crass_ai

phrase_relatedness

penguins_in_a_table

mathematical_induction

arithmetic/3_digit_subtraction

cs_algorithms/valid_parentheses

goal_step_wikihow/goal_inference

play_dialog_same_or_different

cause_and_effect/two_sentences

english_proverbs

identify_odd_metaphor

presuppositions_as_nl

physics

arithmetic/2_digit_multiplication

empirical_judgments

goal_step_wikihow/step_inference

truthful_qa

implicatures

disambiguation_qa

epistemic_reasoning

irony_identification

nonsense_words_grammar

navigate

logical_args

common_morpheme

color/rgb

evaluating_information_essentiality

logical_deduction/three_objects

metaphor_boolean

crash_blossom

fantasy_reasoning

figure_of_speech_detection

logic_grid_puzzle

Examples Alone (no explanations)

`arithmetic/3_digit_division:`
Question: "What is 688 divided by 1?"
Answer: "688"

(choices omitted for brevity)

`identify_odd_metaphor:`
Question: "Which of the following sentences relating to ideas does not use metaphorical language that could also be applied to people? choice: He breathed new life into that idea. choice: It is important how you package your ideas. choice: Cognitive psychology is still in its infancy. choice: That's an idea that ought to be resurrected."
Answer: "It is important how you package your ideas."

Answer Likelihoods

`arithmetic/3_digit_division:`

Question: "What is 688 divided by 1?"
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(choices omitted for brevity)

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Answer: "It is important how you package your ideas."

Each experiment evaluates the relative probability of **this sequence** of tokens relative to all sequences in the **multiple choice answers** (**not** all possible sequences of tokens)

i.e., model's likelihood of **each answer option** after conditioning on the prompt and question

Adding Explanations

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Adding Explanations

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Question: "What is 688 divided by 1?"
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Explanation: "Dividing a number by 1 always yields the same number."

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Answer: "It is important how you package your ideas."
Explanation: 'Packaging does not apply to people, while the other metaphors (breathing life, infancy, and resurrection) do.'"

Adding Explanations

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Answer: "It is important how you package your ideas."
Explanation: 'Packaging does not apply to people, while the other metaphors (breathing life, infancy, and resurrection) do.'"

Does the **likelihood ratio** of the correct answer **increase** for the **next question**, **when explanations are added** to previous examples?

How much do explanations help?

Is it <u>the explanation itself</u> or something else (relevant words, syntactic structure, ...)

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Answer: "It is important how you package your ideas."
Explanation: 'Packaging does not apply to people, while the other metaphors (breathing life, infancy, and resurrection) do.'"

vs 3 Control Conditions

Control Experiments

True Non-Explanation	**Other Item Explanation**	**Scrambled Explanation**

How much do explanations help?

Is it <u>the explanation itself</u> or something else (relevant words, syntactic structure, ...)

`arithmetic/3_digit_division`:
Question: "What is 688 divided by 1?"
Answer: "688"
Explanation: "Dividing a number by 1 always yields the same number."
True non-explanation: "688 is an even number, which means it is divisible by 2."

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True non-explanation: "Cognitive psychology involves the study of people, and sometimes comparative study of other animals to determine the similarities and differences."

Control Experiment 1

Relevant but non-explanatory

How much do explanations help?

Is it the explanation itself or something else (relevant words, syntactic structure, ...)

Control Experiment 2

`arithmetic/3_digit_division`:
Question: “What is 688 divided by 1?”
Answer: “688”
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Other item explanation: “We know 450 = 9 * 50, so we can rewrite as 522 / 9 = 450 / 9 + 72 / 9 = 50 + 8 = 58.”

`identify_odd_metaphor`:
Question: “Which of the following sentences relating to ideas does not use metaphorical language that could also be applied to people? choice: He breathed new life into that idea. choice: It is important how you package your ideas. choice: Cognitive psychology is still in its infancy. choice: That’s an idea that ought to be resurrected.”
Answer: “It is important how you package your ideas.”
Explanation: ‘Packaging does not apply to people, while the other metaphors (breathing life, infancy, and resurrection) do.‘”
True non-explanation: “Cognitive psychology involves the study of people, and sometimes comparative study of other animals to determine the similarities and differences.”
Other item explanation: “This sentence does not use a metaphor, while the others use container-relevant metaphors (fullest, crammed, contained).”

Randomly chosen from other examples in the prompt dataset

How much do explanations help?

Is it <u>the explanation itself</u> or something else (relevant words, syntactic structure, …)

Control Experiment 3

`arithmetic/3_digit_division`:
Question: "What is 688 divided by 1?"
Answer: "688"
Explanation: "Dividing a number by 1 always yields the same number."
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Scrambled explanation: "number. the Dividing same always 1 yields a number by"

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These are **untuned explanations:**

Annotated by hand, **without looking at LM probabilities**

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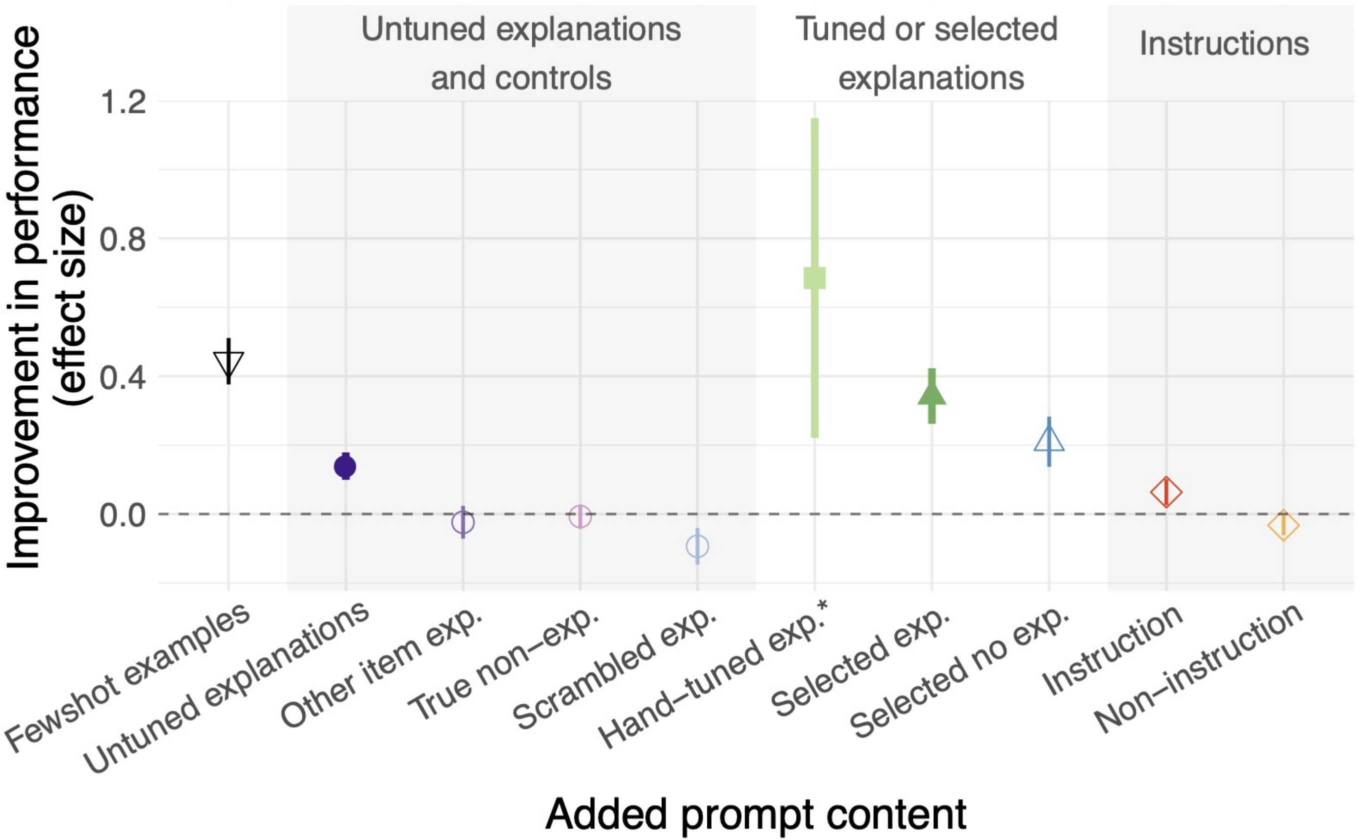
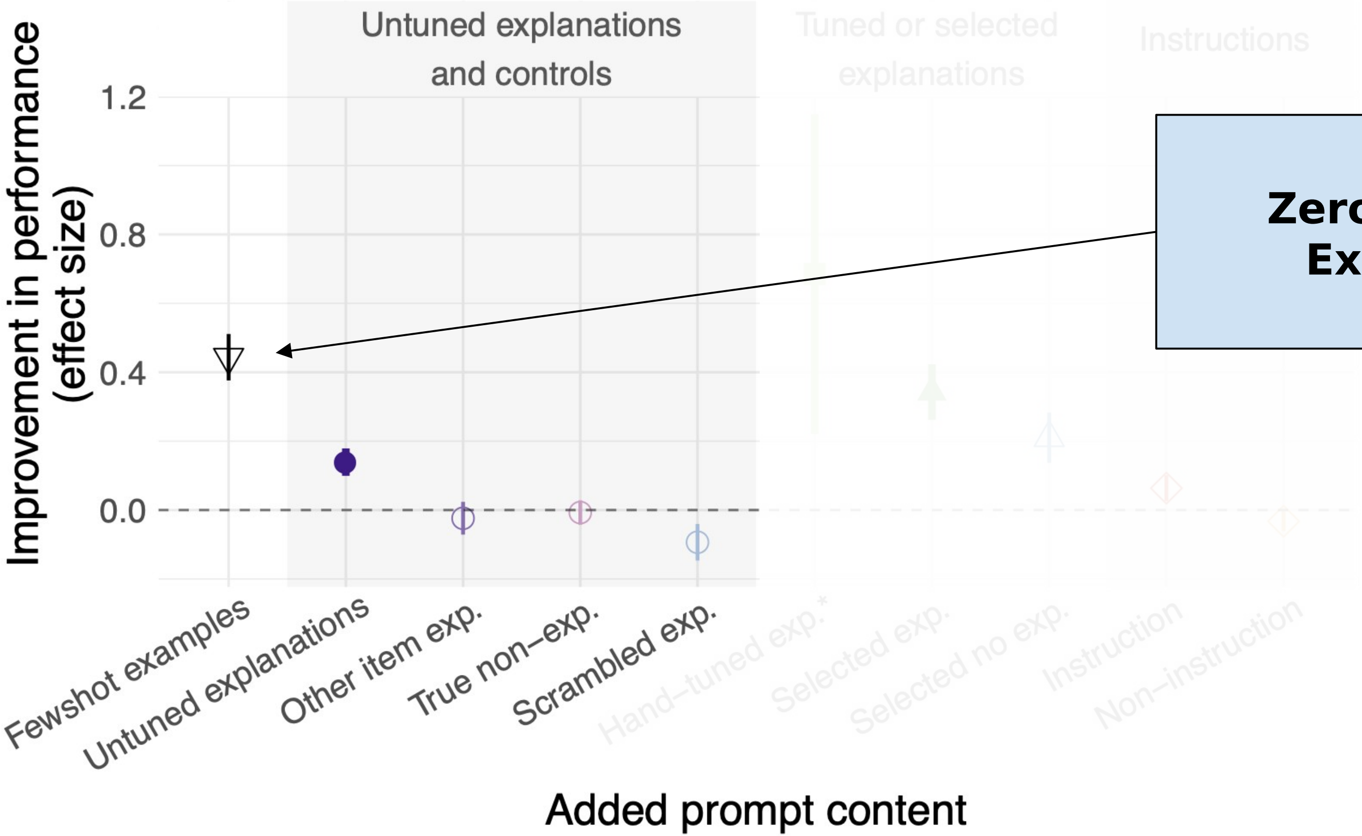
Hierarchical Regression: **quantify** unique added contribution (on Gopher-280B)

Effect Size

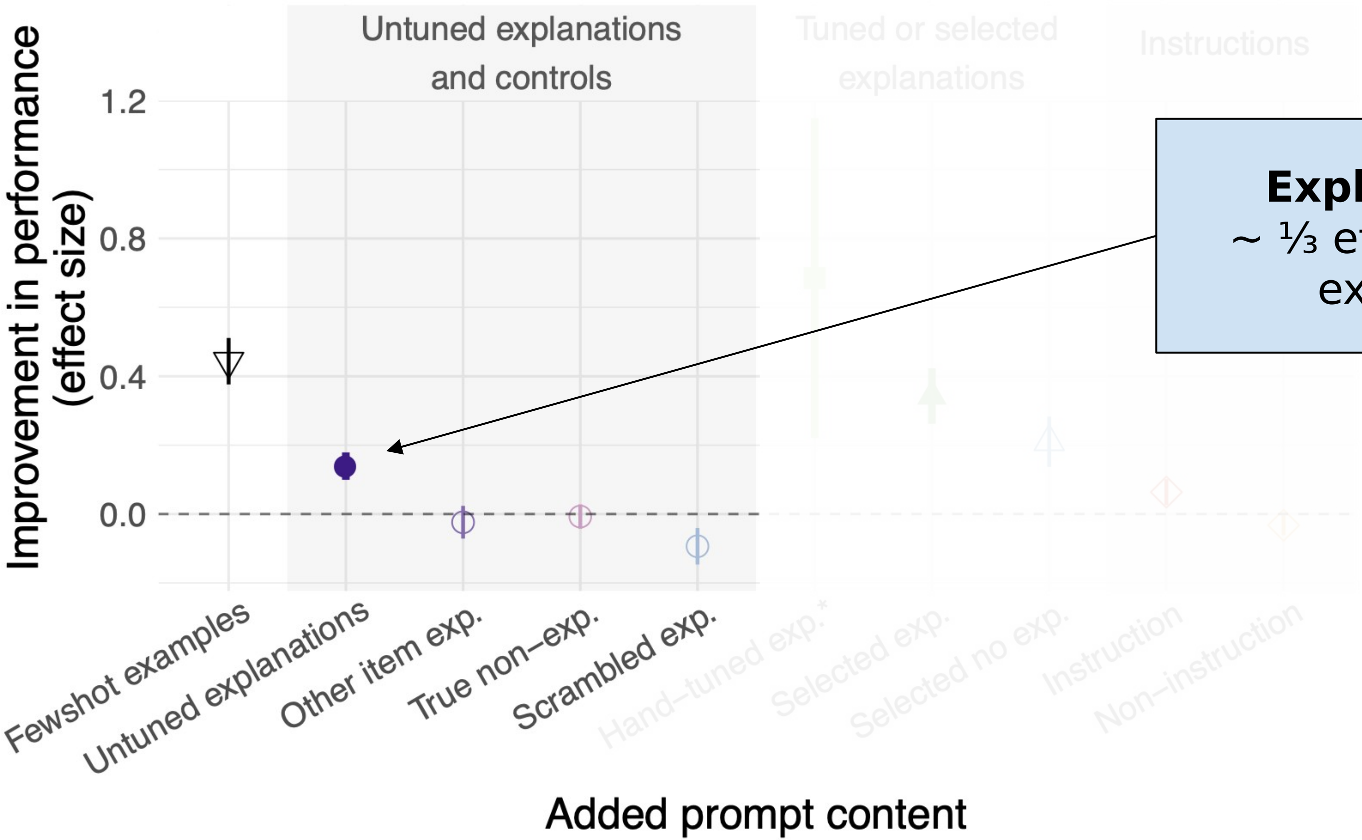


Effect Size



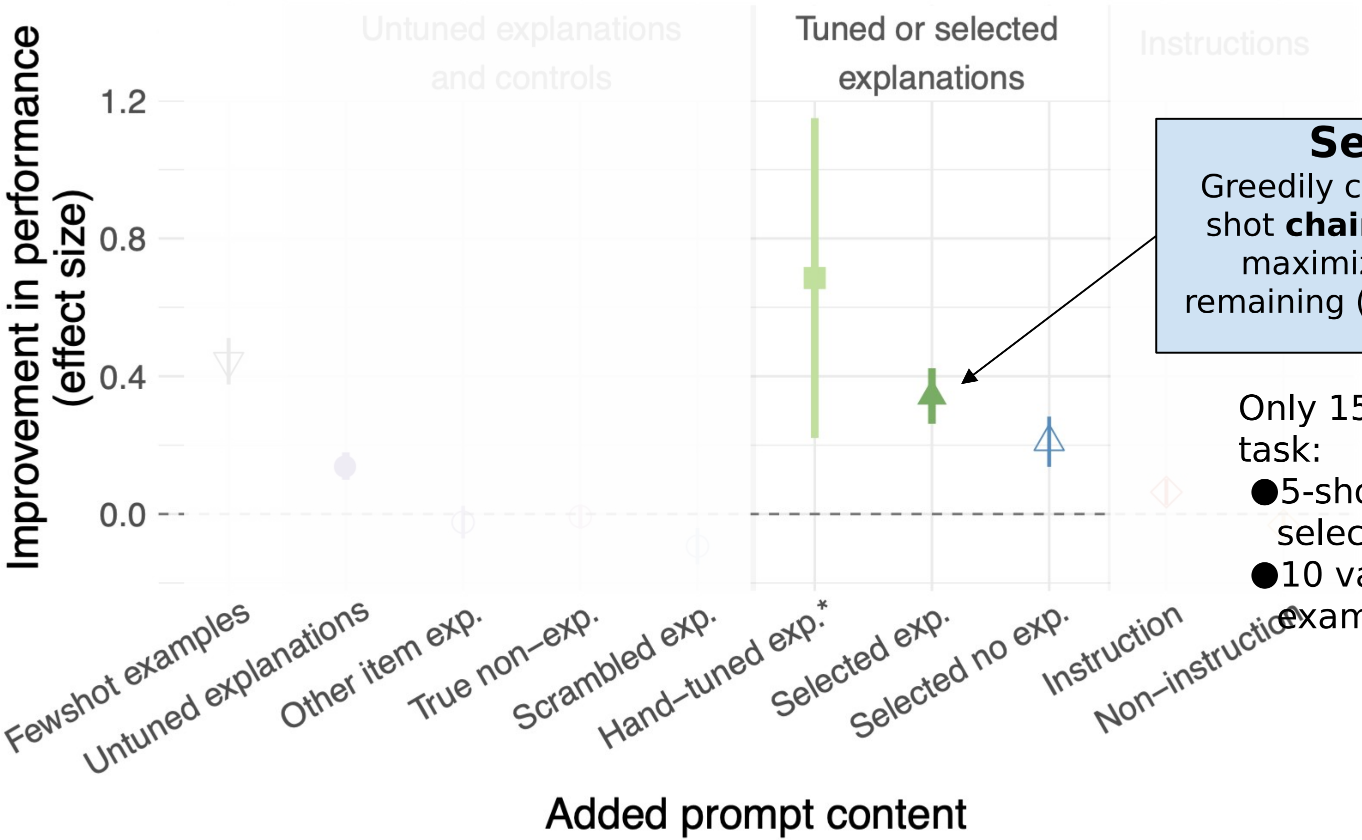
Zero-Shot vs Examples

Effect Size



Explanations
~ ⅓ effect size vs examples

Effect Size



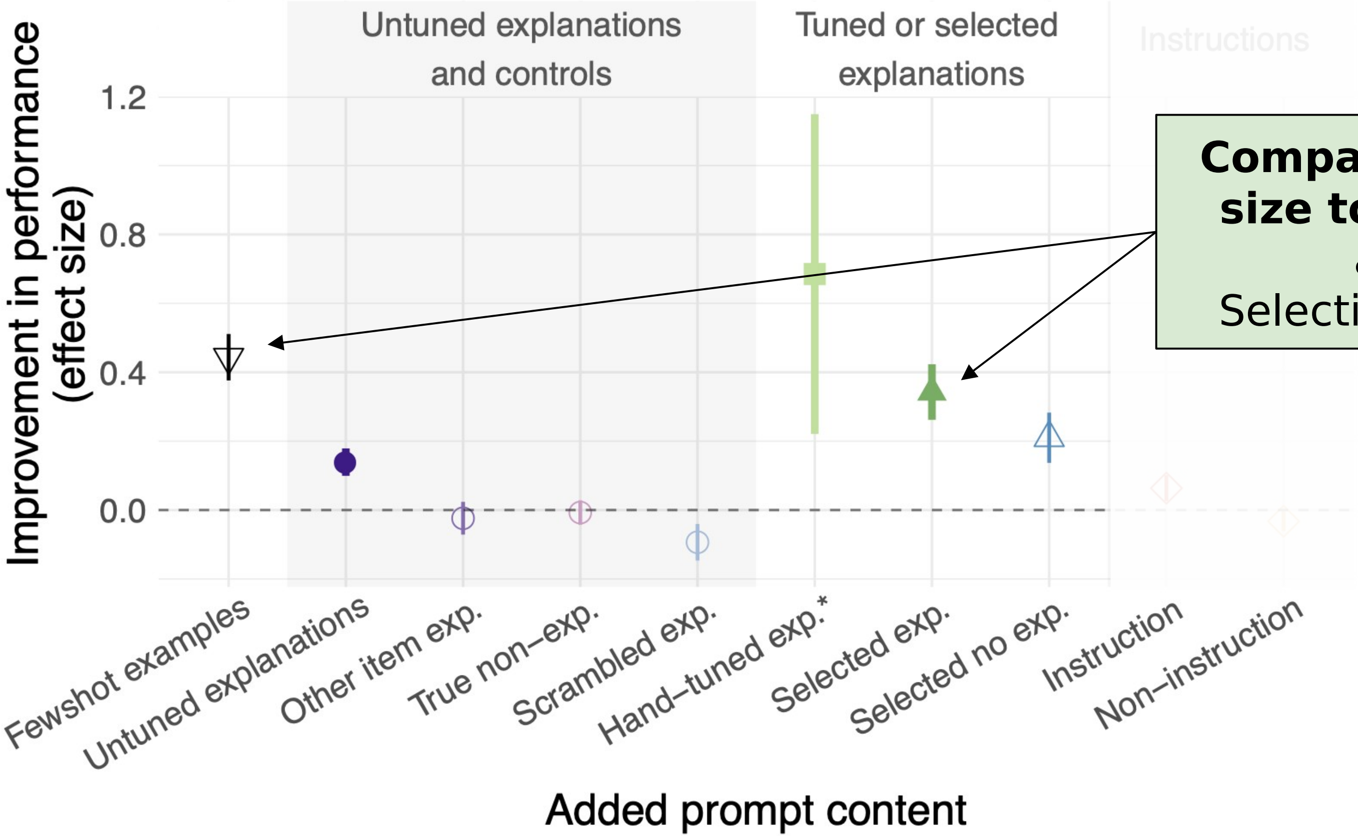
Selection

Greedily choose the best k-shot **chain of prompts** to maximize likelihood of remaining (n-k) examples, up to k=5

Only 15 examples per task:

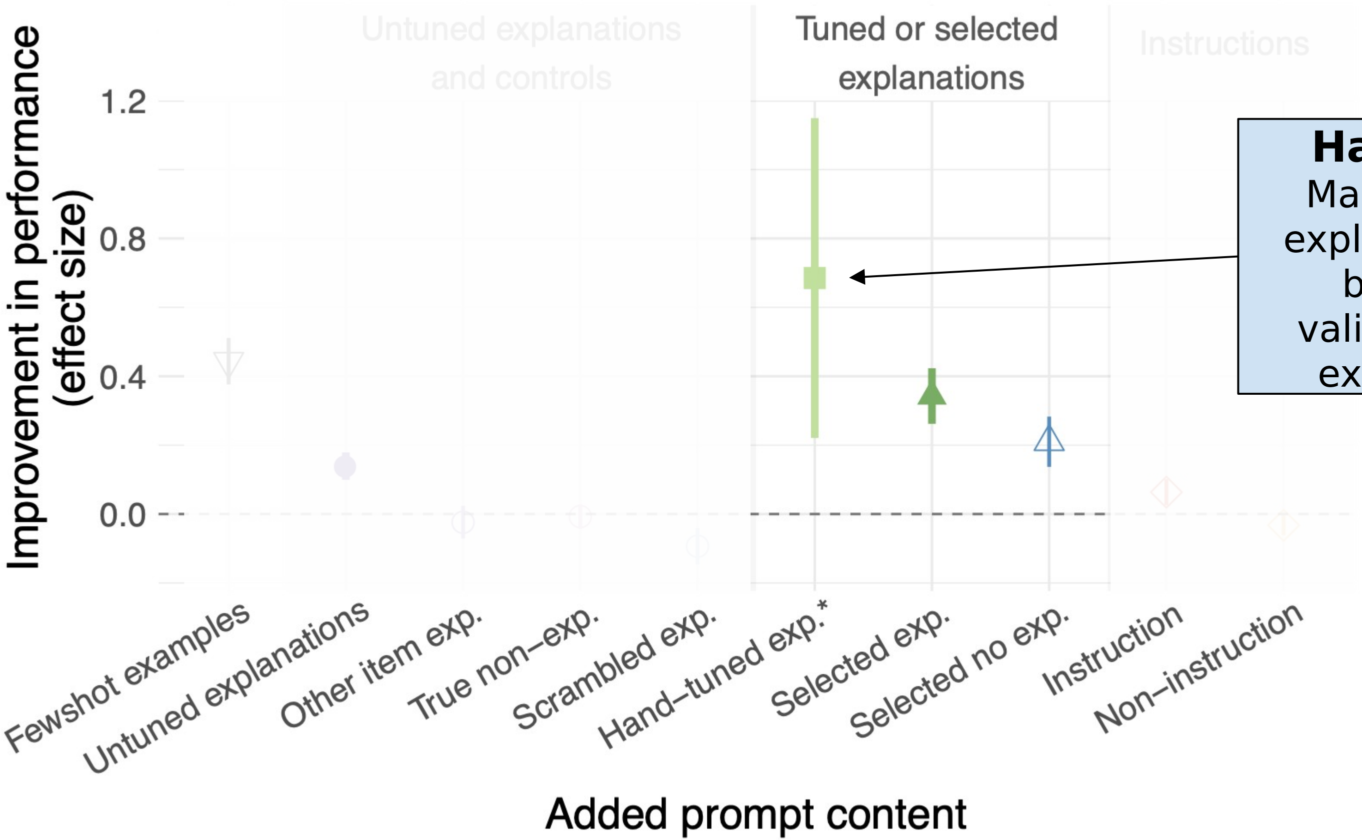
- 5-shot prompt selection
- 10 validation examples

Effect Size



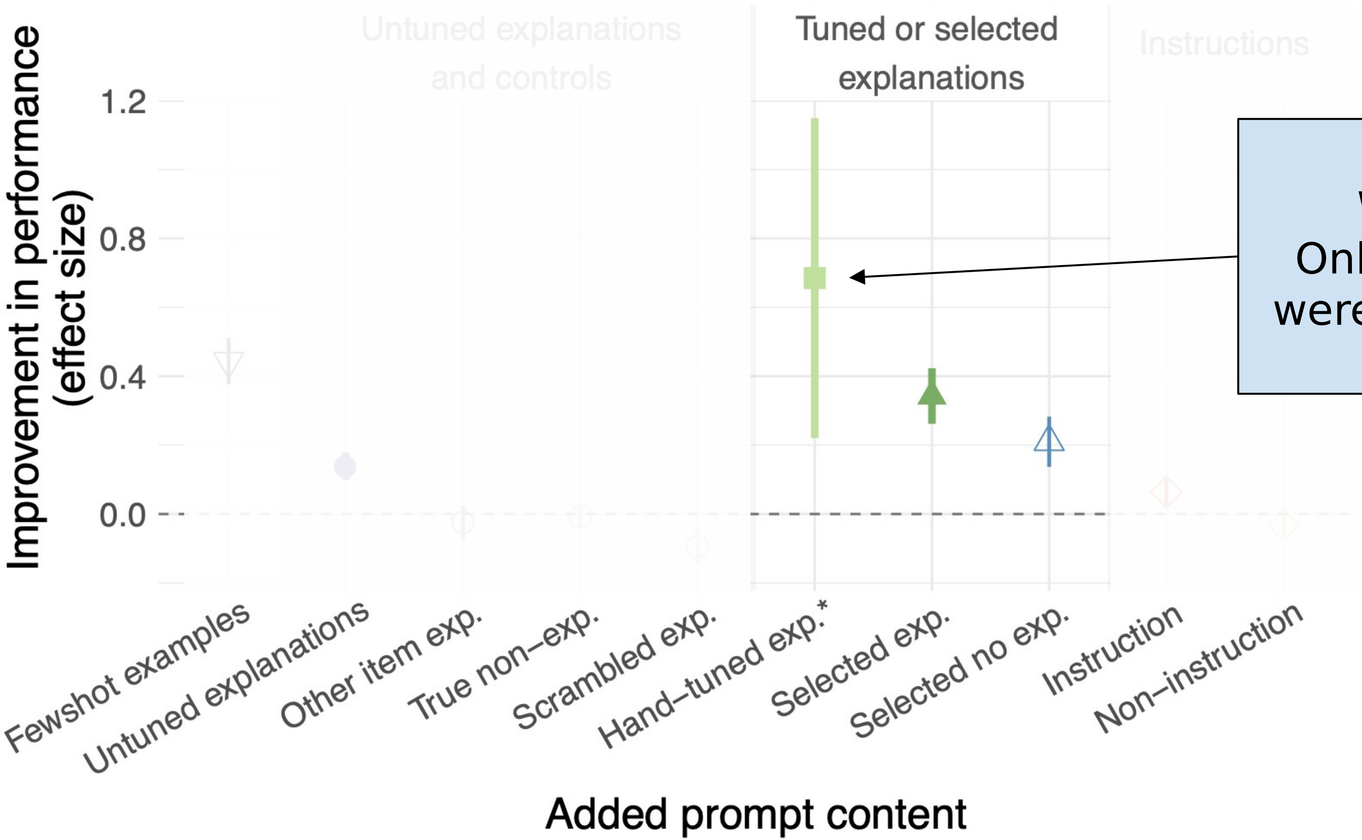
Comparable effect size to examples alone
Selection is useful!

Effect Size



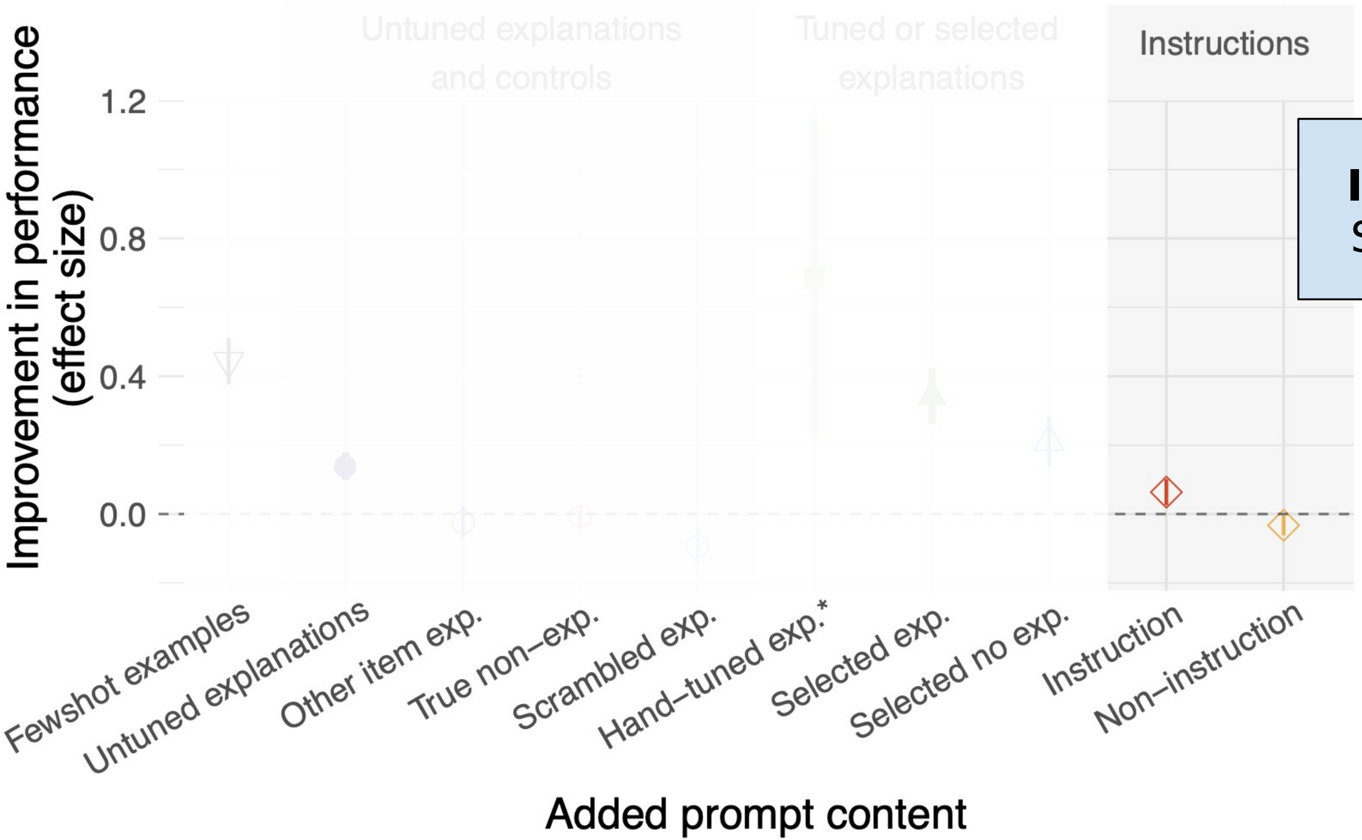
Hand-tuning
Manually reword explanation text to boost acc on validation set (10 examples/task)

Effect Size



Wide CI?
Only 5/40 tasks were hand-tuned

Effect Size



Instructions
Small benefit

Presentation Roadmap

- Introduction
- Method
- **Experiments**
- Discussion

Potential impacts for LLM applications

- Adding untuned few-shot explanations slightly outperforms examples alone
- Selection and rewriting using a **validation set** boosts gains as much as examples
- **In-context** explanations vs finetuning LLM **weights** on a task
    - Explanations are quicker + easier to deploy
    - But ha
    - Can th

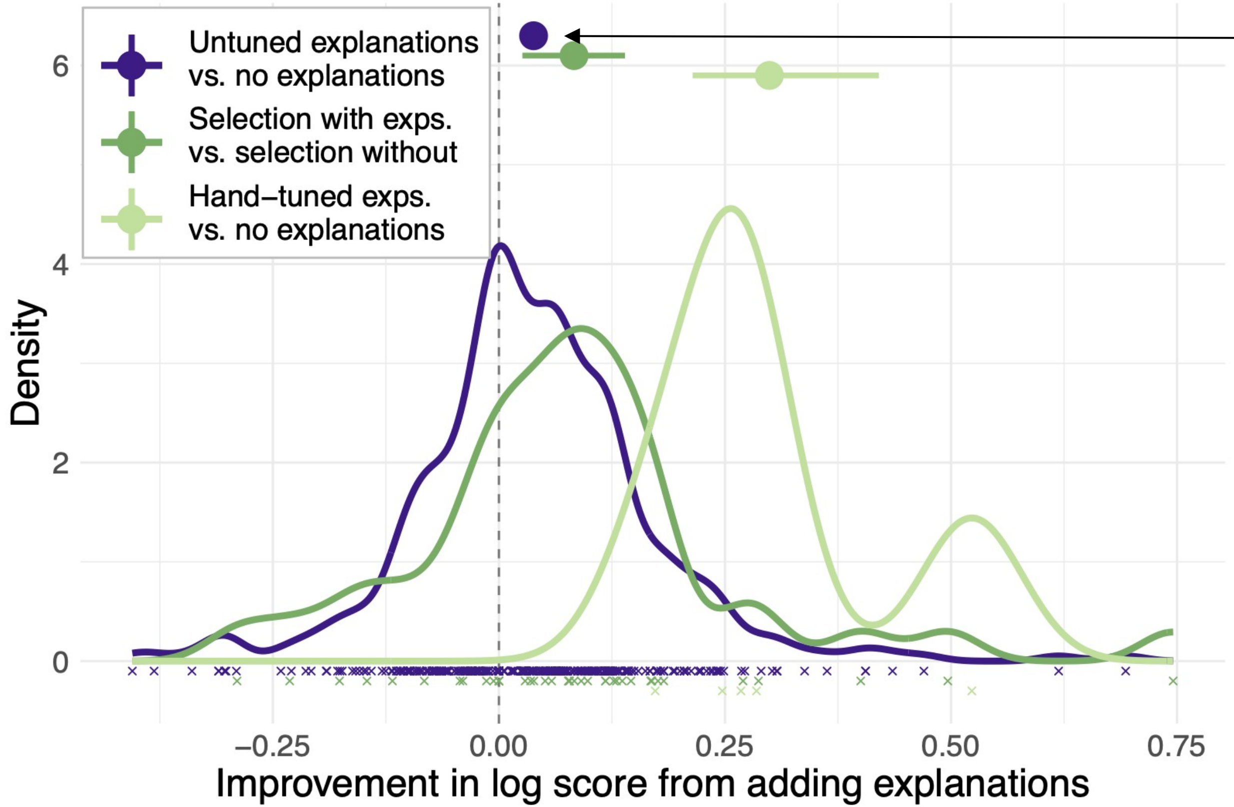
What is the “budget” for in-context learning?

**Gopher:
2048 tokens**

**GPT-4-32k
~50pgs of text**

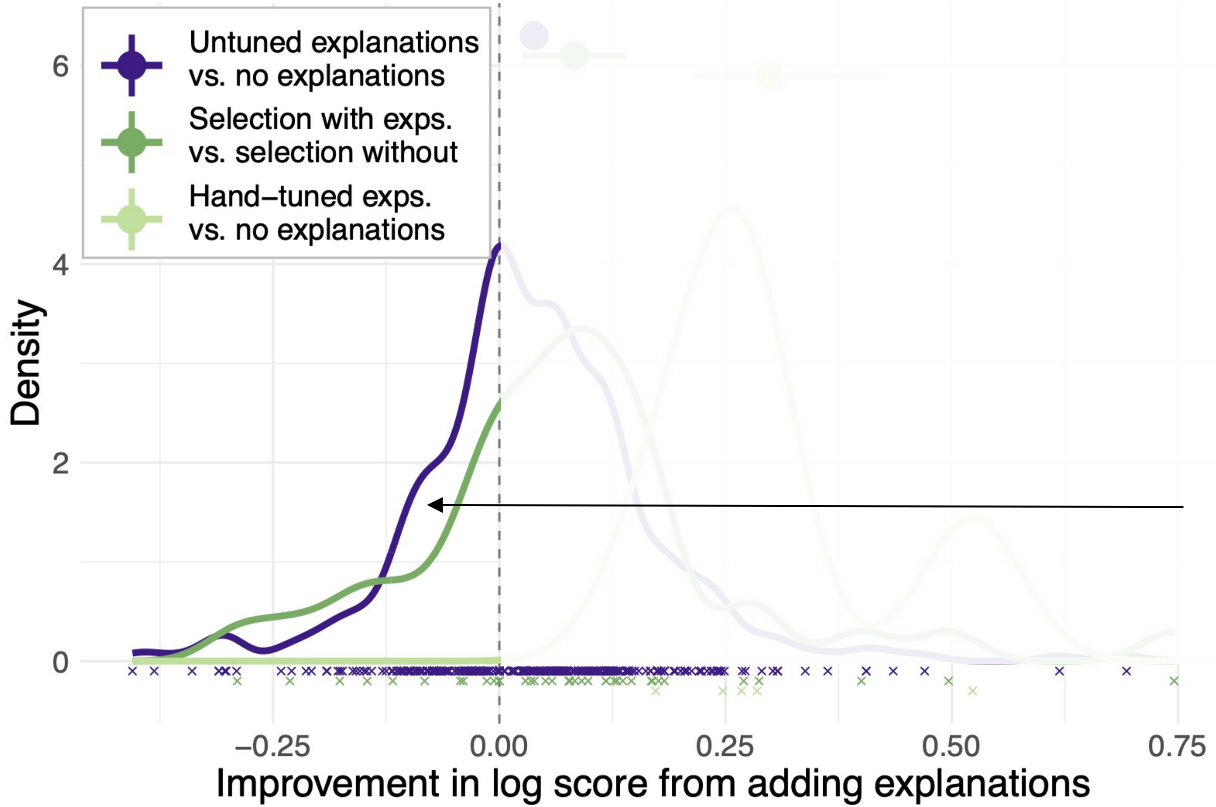
Source:
https://openai.com/research/gpt-4

Selection vs untuned explanations



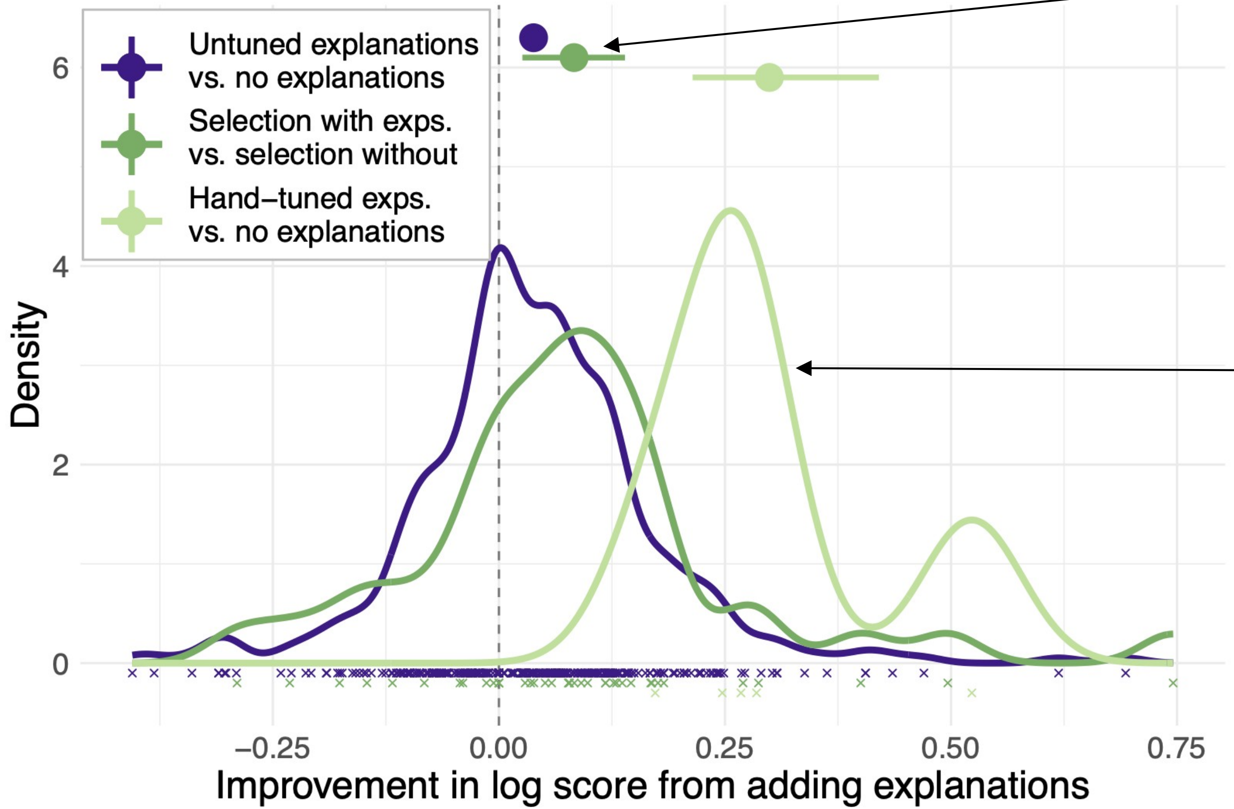
Untuned explanations are slightly beneficial on average

Selection vs untuned explanations



But many instances where untuned explanations hurt accuracy

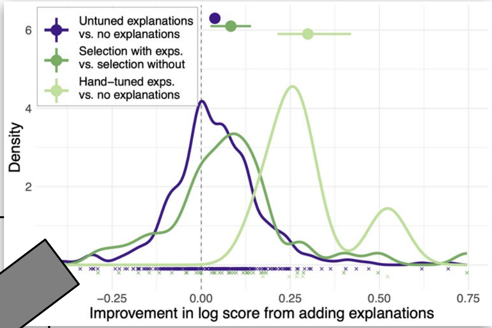
Selection vs untuned explanations

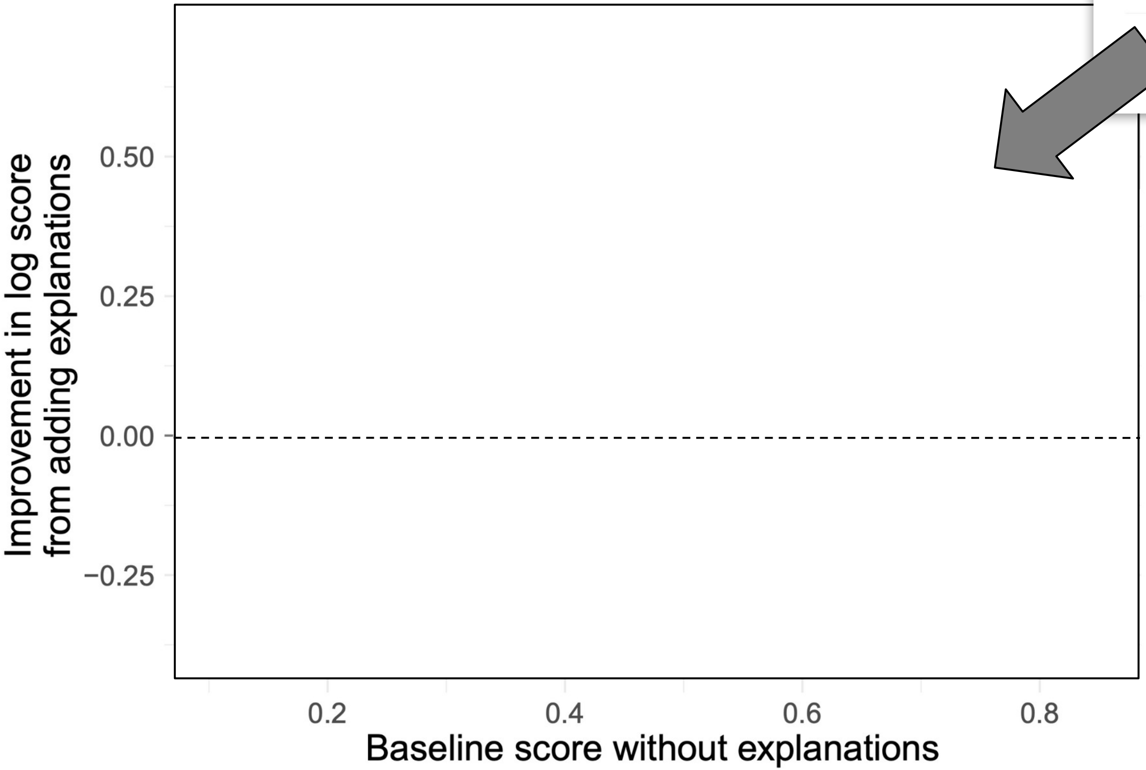


Benefit of selection is clearer

Not enough data on hand-tuning to be persuasive?

When do explanations help?

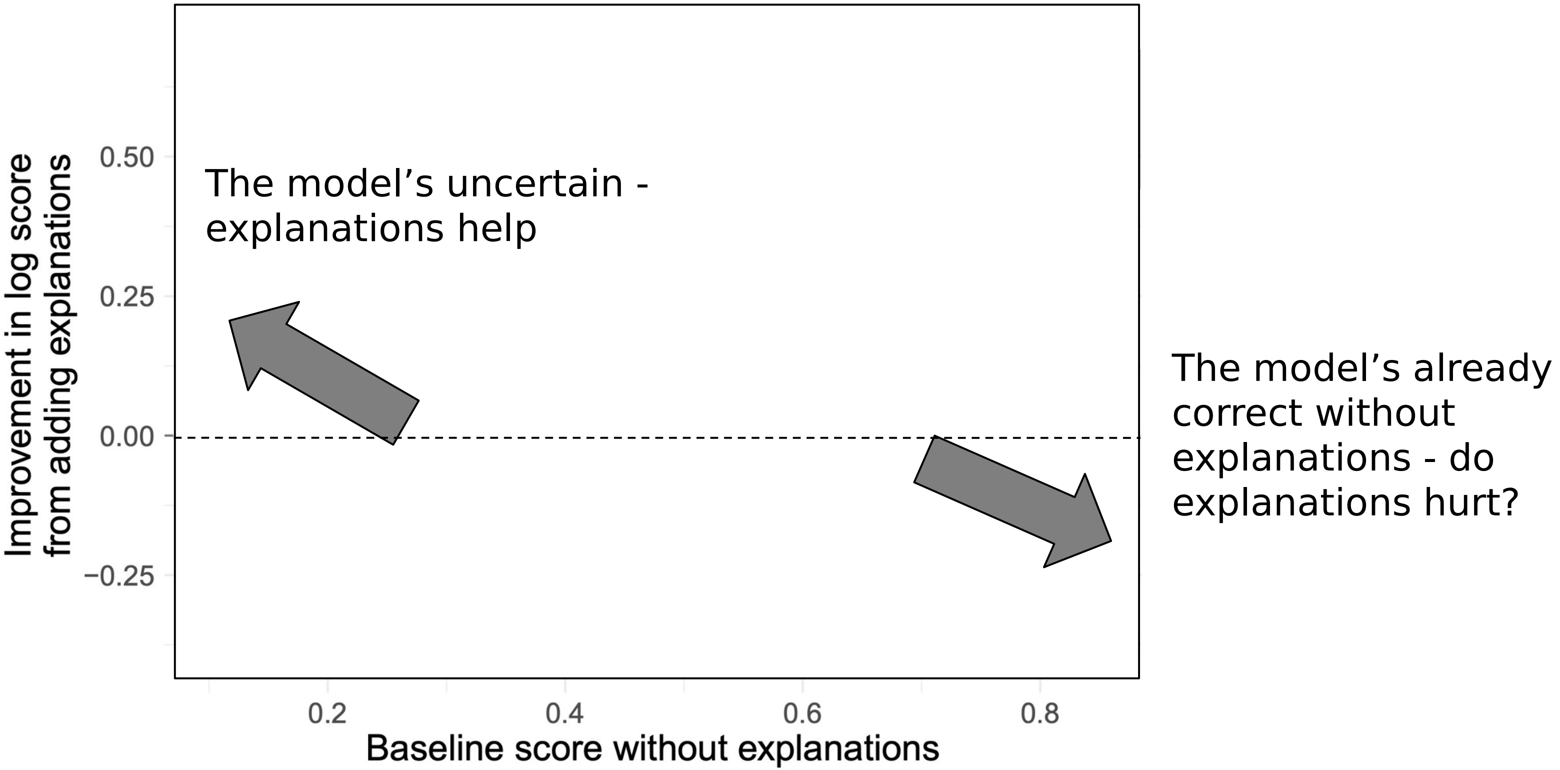




Addition of explanations vs model’s **prior** confidence **with examples alone**

Baseline: without explanations (baseline is **not** zero-shot)

When do explanations help?



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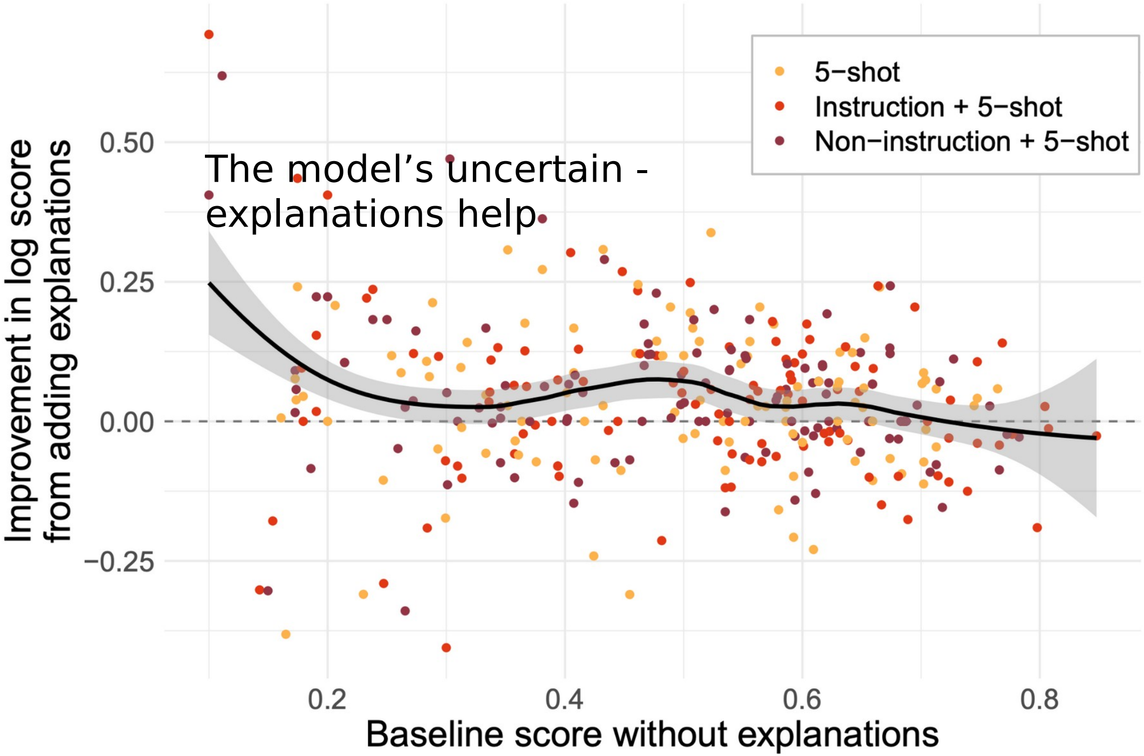
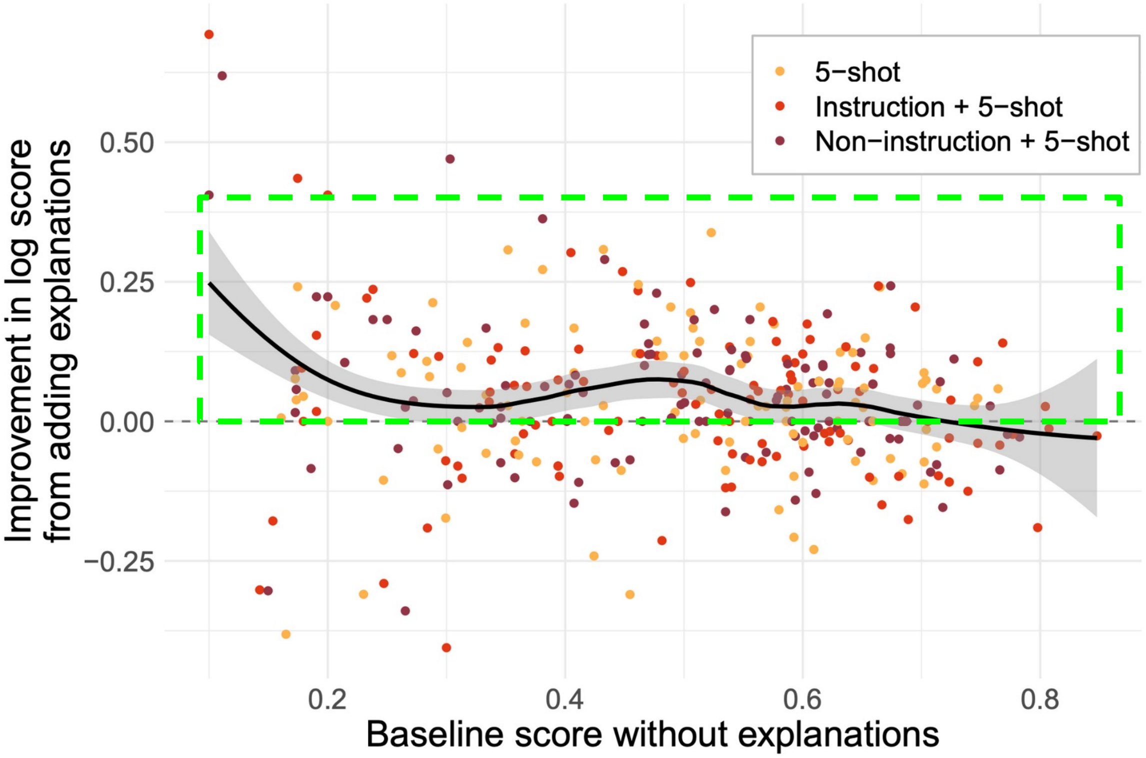


The model’s uncertain - explanations help

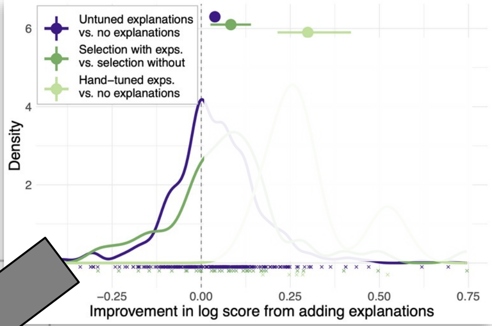
The model’s already correct without explanations - do explanations hurt?

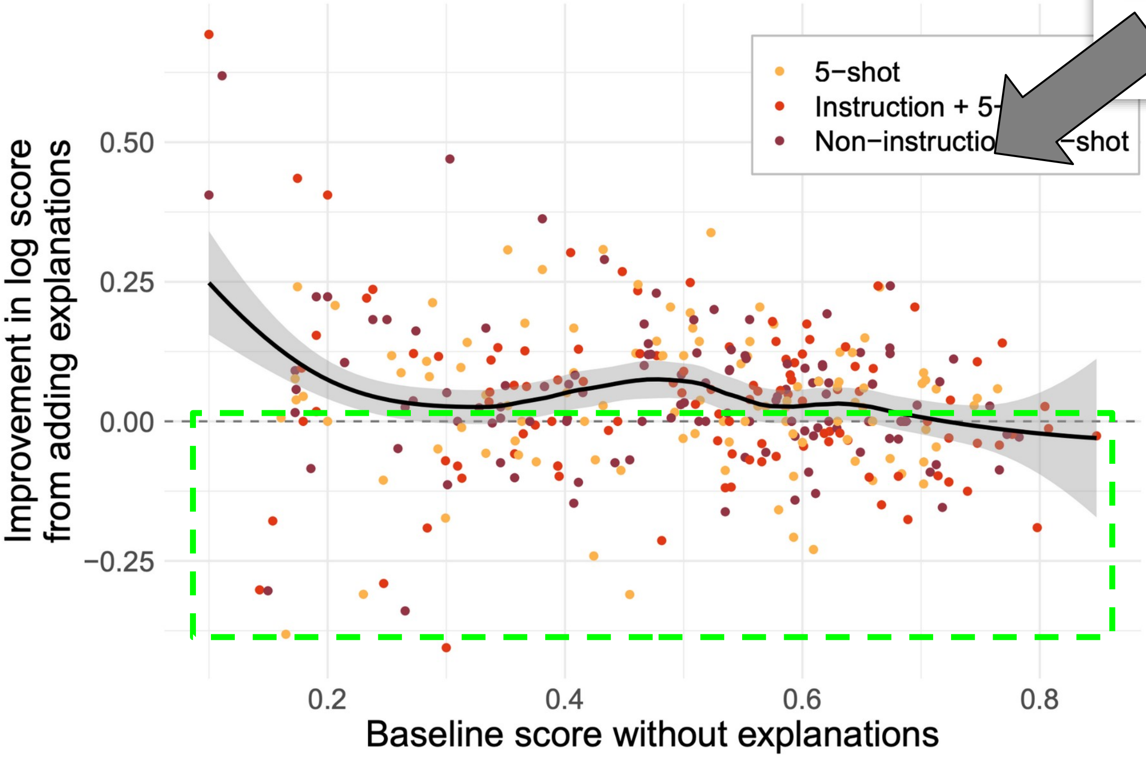
When do explanations help?



Explanations usually help: higher density regardless of baseline confidence

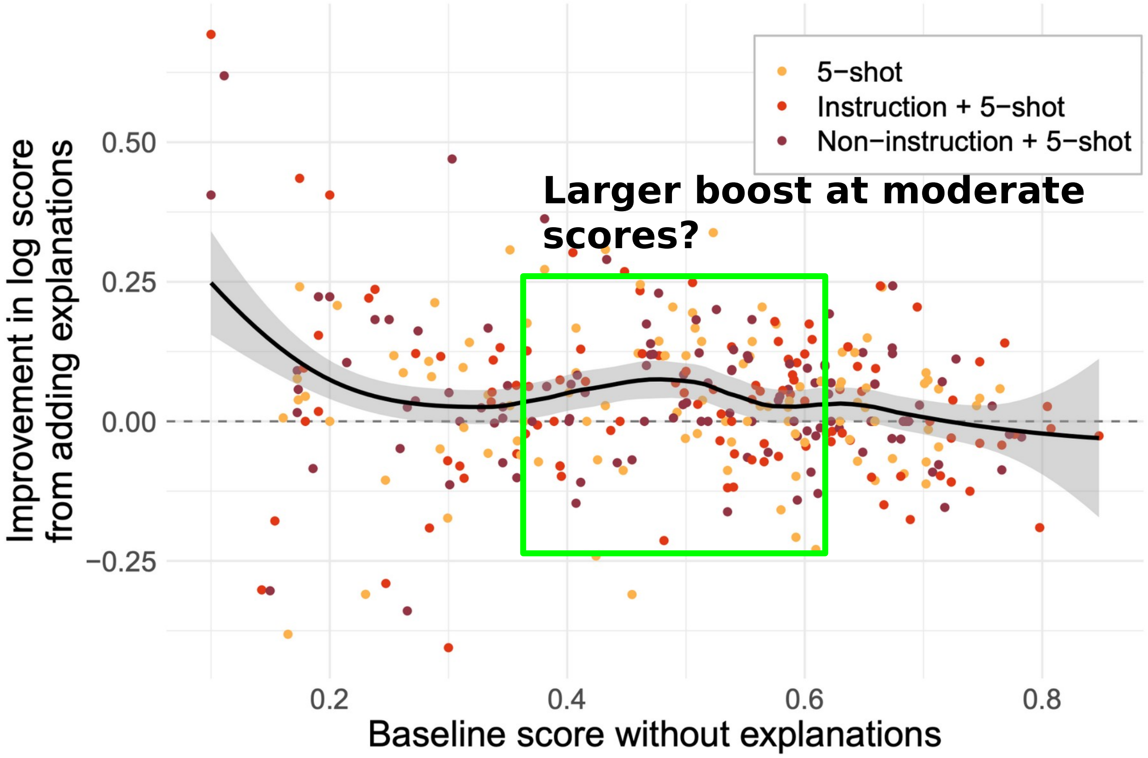
When do explanations help?





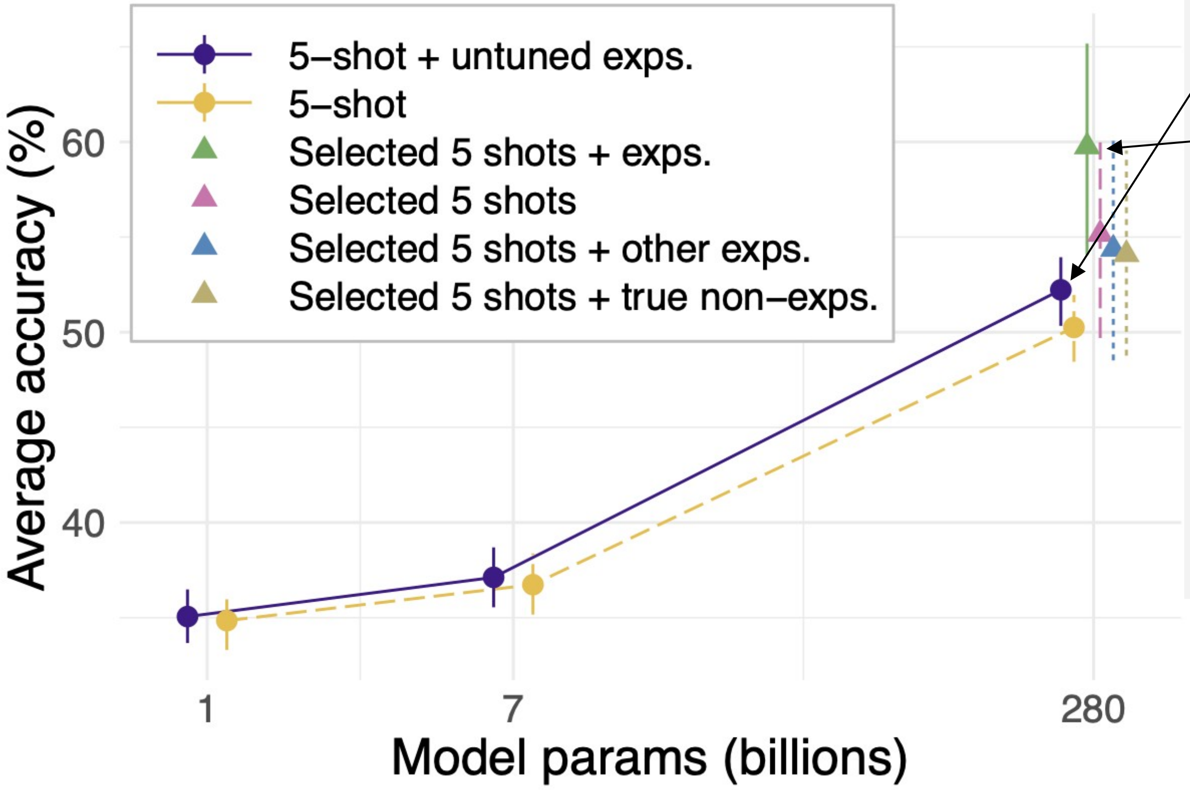
But not always: scores also drop with explanations, **but uniformly(ish)**

When do explanations help?



“**Zone of proximal effect**?” (what a learner can do **on the cusp of their prior capabilities,** with/without teacher assistance)

Is Scale Necessary?



- Untuned explanations only help modestly **for the largest model**
  - **Selection** emerges from the pack
- But **no benefits @ 1B, 7B**
- Consistent w/prior work on scaling:
  - Brown ‘20: **sharp transition** for arithmetic
  - Wei ‘21: only large LMs generalize

Task Specific Improvements

Does **improvement** provided by explanations **differ** between **types** of **tasks**?

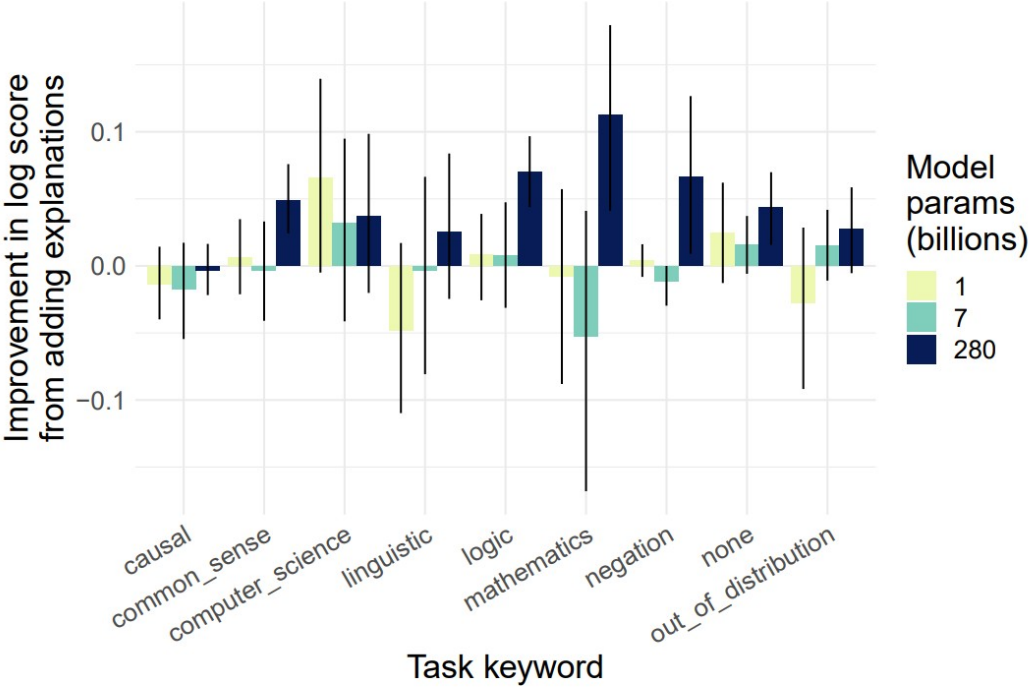
Task Diversity

analytic_entailment
odd_one_out
physical_intuition
unit_interpretation/lv2
riddle_sense
arithmetic/3_digit_division
crass_ai
phrase_relatedness
penguins_in_a_table
mathematical_induction
arithmetic/3_digit_subtraction
cs_algorithms/valid_parentheses
goal_step_wikihow/goal_inference
play_dialog_same_or_different
cause_and_effect/two_sentences
english_proverbs
identify_odd_metaphor
presuppositions_as_nl
physics
arithmetic/2_digit_multiplication
empirical_judgments
goal_step_wikihow/step_inference
truthful_qa
implicatures
disambiguation_qa
epistemic_reasoning
irony_identification
nonsense_words_grammar
navigate
logical_args
common_morpheme
color/rgb
evaluating_information_essentiality
logical_deduction/three_objects
metaphor_boolean
crash_blossom
fantasy_reasoning
figure_of_speech_detection
logic_grid_puzzle

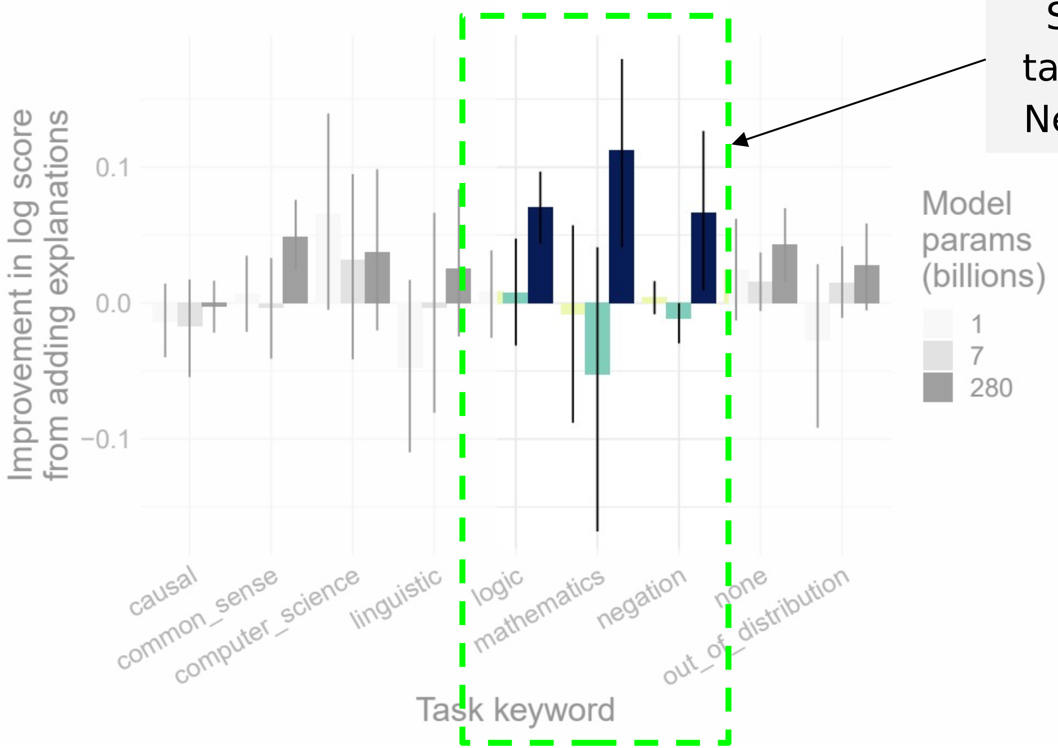
Clustering Tasks into Categories

- Casual
- Computer Science
- Logic
- Negation
- OOD
- Common Sense
- Linguistic
- Mathematics
- None

Not all Tasks are Created Equal



Not all Tasks are Created Equal



Semantics of explanations for tasks about Logic, Mathematics, Negation might be better suited for LM

Presentation Roadmap

- Introduction
- Method
- Experiments
- Discussion

Discussion

- Can explanations improve few-shot
- Why are post-answer explanations
- What do their results imply about LMs’ abilities for in-
- How do explanations relate to
- How does their work relate to human language processing?

Discussion

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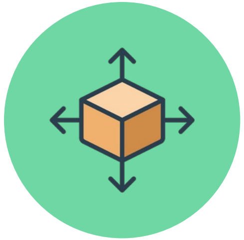
Can explanations improve few-shot learning?



Scale

Quality

Discussion

- Can explanations improve few-shot
- Why are post-answer explanations
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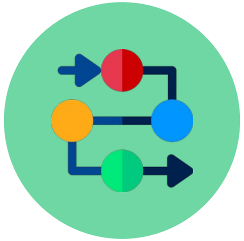
Why are post-answer explanations interesting?



Holistic vs Chain-of-Reasoning

Test Time Implications

Discussion

- Can explanations improve few-shot
- Why are post-answer explanations
- What do their results imply about LMs' abilities for in-
- How do explanations relate to
- How does their work relate to human language processing?

What do their results imply about LMs’ abilities for in-context learning?

Example & Explanation Relationships

Learning vs Recalling

Discussion

- Can explanations improve few-shot
- Why are post-answer explanations
- What do their results imply about LMs’ abilities for in-
- How do explanations relate to
- How does their work relate to human language processing?

How do explanations relate to instructions?

Instructions

Improvement in Prior Work was Successful

Instruction & Explanations

Complementary

Discussion

- Can explanations improve few-shot
- Why are post-answer explanations
- What do their results imply about LMs' abilities for in-
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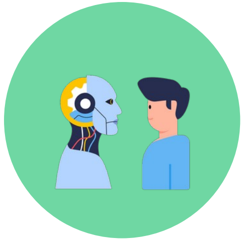
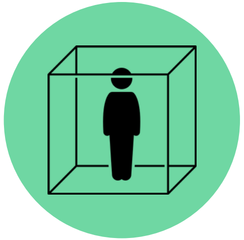
- How does their work relate to human language processing?



Humans ≠ LMs



No Broader Context

Class Discussion

- Is the dataset large (~600 samples)/diverse (40 tasks) enough to support their conclusions?
- Is the degree of improvement offered by explanations significant?
- How does this compare to other **in-context learning** methods?
  - Chain-of-thought [Wei]
  - Soft prompts (changing the embedding of the prompt) [Leister]
  - Finetuning + Distillation (Alpaca) [Taori]
- How could explanations benefit smaller models and not only large models?